

WATER TRAVEL IN PLANTS

The Celery Experiment *Detailed Lesson Plan*

Subject	Environmental Science / General Science
Topic	Water Travel in Plants — How does water move from roots to leaves?
Duration	60 Minutes (can be split into 2 × 30 min sessions)
Format	Story-based Introduction → Concept Teaching → Hands-on Experiment → Observation → Recap
Board	CBSE / ICSE / State Board Compatible

◆ SECTION 1 — LESSON OVERVIEW

This lesson introduces students to the concept of water transport in plants through an engaging narrative, concept explanation with Indian plant examples, and a vivid hands-on experiment using celery and coloured water. Students will observe how water travels from roots through the stem to the leaves — making an invisible process visible and memorable.

Big Question for the Lesson:

"How did the water get from the soil all the way to the top leaves?"

◆ SECTION 2 — LEARNING OBJECTIVES

By the end of this lesson, students will be able to:

LO 1	REMEMBER	Name the three main parts of a plant involved in water travel: roots, stem, and leaves.
LO 2	UNDERSTAND	Explain in simple words how water moves from soil → roots → stem → leaves.
LO 3	APPLY	Conduct the celery experiment step-by-step and record observations correctly.
LO 4	ANALYSE	Connect the colour change in celery leaves to the concept of water-carrying tubes in the stem.
LO 5	EVALUATE	Predict what might happen with other plants (tulsi, neem) based on what they observed.

◆ SECTION 3 — KEY VOCABULARY & CONCEPTS

Roots	The underground part of a plant that soaks up water and nutrients from the soil. Like a drinking straw in the ground!
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Stem	The part that connects roots to leaves. It contains tiny tubes that carry water upward. Like a water highway!
Leaves	The flat green parts that receive water, stay fresh, and make food for the plant using sunlight.
Absorb	To soak up or take in. Roots absorb water from the soil.
Tubes / Xylem	Tiny straw-like tubes inside the stem that carry water from roots all the way to the leaves.
Observe	To look carefully and notice what is happening. Scientists always observe!
Experiment	A test we do to find the answer to a question. Today's experiment uses celery and coloured water.
Transport	To carry or move something from one place to another. Water is transported through the plant.

Hindi / Indian Language Support: Roots = Jaad (जाद), Stem = Tana (तना), Leaves = Patte (पत्ते), Water = Paani (पानी)

◆ SECTION 4 — SESSION OUTLINE WITH TIMING

0–5 min	Hook — Story Warm-Up Read the Tulsi story aloud. Ask: 'Have you ever forgotten to water a plant? What happened?' Build curiosity around the big question.
5–12 min	Concept 1 — Why Do Plants Need Water? Explain 3 reasons with Indian examples. Use call-and-response: 'Water helps plants grow ____!' (students say 'strong and green!')
12–18 min	Concept 2 — The Journey: Roots → Stem → Leaves Teach the three-part journey. Use the 'drinking straw' and 'sugarcane juice' analogies. Draw on board.
18–22 min	Indian Plants — Making it Real Show/name 7 Indian plants (Tulsi, Dhaniya, Ganna, Palak, Neem, Marigold, Money Plant). Ask students to name plants at home.
22–28 min	Experiment Setup — Materials & Steps 1–2 Distribute materials. Teacher demonstrates filling the glass and adding food colour. Students prepare their own setup.
28–35 min	Experiment Steps 3–4 — Place Stalk & Wait Students place celery stalks in coloured water. Set a 30-min check timer. Begin observation worksheet while waiting.
35–45 min	Observation & Discussion Check celery every 10 minutes. Discuss: What changed? Where did the colour go? Why? Use the 3 discussion questions from slides.
45–52 min	Worksheet & Drawing Activity Students complete the observation worksheet: draw BEFORE/AFTER celery and fill in the sentence strip.
52–58 min	Recap — Key Words & What Did We Learn? Review 4 key words (Roots, Stem, Leaves, Absorb/Observe). Use 3-point recap: need water → roots to stem → experiment proves it.
58–60 min	Home Extension Assign: Look at any plant at home tonight. Try with white mogra flowers. Draw what you see.

◆ SECTION 5 — HOOK STORY (Read Aloud)

WHY DID THE TULSI PLANT LOOK SAD?

Arjun forgot to water the tulsi plant for two days. The plant was droopy and sad. His Nani came outside and said, "The plant is thirsty, beta!" Arjun quickly got a glass of water and poured it into the soil. By evening, the tulsi plant stood tall and fresh again — its leaves were bright and green!

Discussion Question for Class: How did the water get from the soil all the way to the top leaves? Let's find out!

Teacher Tips for Storytelling:

- Use a dramatic, slow voice when the plant is 'sad' and a bright, happy voice when it recovers.
- Pause before the question and let 2–3 students guess before moving on.
- Hold up or point to a real tulsi plant if available in the classroom.

◆ SECTION 6 — CONCEPT EXPLANATIONS

6A — Why Do Plants Need Water?

Grows Strong

Water helps plants grow strong and green! Without it, leaves droop and the plant becomes weak.

Gives Energy

Water gives plants energy and freshness — just like how we feel better after drinking water on a hot day!

Like Our Drinks

Water is to plants like chai or milk is to us in the morning — they cannot start the day without it!

6B — Where Does Water Enter? (The Roots)

Roots are underground — they grow down into the soil to find water and food for the plant. Think of them like a bundle of drinking straws buried under the ground! When you water a plant, the roots soak up the water from the soil and start sending it upward.

Indian Example: Next time Amma or Dadi pulls out coriander (dhaniya) or spinach (palak) from the kitchen garden, look at the roots — you can see the thin, thread-like roots that absorb water!

6C — How Does Water Travel Up? (The Stem)

Inside the stem, there are **hundreds of tiny tubes** — so small you cannot see them without a microscope. These tubes carry water upward from the roots to the leaves, just like many thin drinking straws bundled together! The stem is the plant's **water highway**.

Indian Analogy: Imagine drinking sugarcane juice (ganna ka ras) through a long straw — the juice travels all the way up from the bottom to your mouth. That is exactly what the stem does for the plant, every single day!

6D — Where Does Water Go? (The Leaves)

Once the water reaches the top of the stem, it flows into the **leaves and flowers**. The leaves use water to stay fresh, make food using sunlight (photosynthesis — a big word for Grade 3!), and release extra water into the air.

Indian Example: Have you noticed flower sellers keeping roses in big water buckets at the bazaar? Or marigolds staying bright and fresh when kept in water? That is water travelling through the stem to the petals, keeping them alive and beautiful!

◆ SECTION 7 — INDIAN PLANTS THAT SHOW WATER TRAVEL

All of these familiar Indian plants carry water from their roots to their leaves every single moment — even right now!

Tulasi (Holy Basil)	Found in almost every Indian home. Its upright stem carries water to the fragrant leaves.
Dhaniya (Coriander)	The thin, delicate stems still carry enough water to keep fresh green leaves bright.
Ganna (Sugarcane)	One of the best examples — a very tall stem (up to 4 metres!) carrying water and sugary sap all the way up.
Palak (Spinach)	Wide, flat leaves need lots of water — the stem works hard to supply them all.
Neem	A large tree whose roots go very deep to find underground water, which then travels up the tall trunk to thousands of leaves.
Marigold (Genda)	Festival favourite! Keep marigolds in water and they stay fresh for pooja or decoration much longer.
Money Plant	A common indoor plant that can even grow in just a glass of water — the stem absorbs water directly!

Teacher Tip: Ask students which plants are in their homes or gardens. Create a quick class list on the board!

◆ SECTION 8 — THE CELERY EXPERIMENT

This experiment makes water travel **visible**. The coloured water travels up through the tiny tubes in the celery stem and reaches the leaves — turning them the colour of the water!

8A — Materials Required (Per Group / Per Student)

● Fresh celery stalk with leaves	Choose stalks with bright green leaves at the top. The leaves show colour change most clearly.
● Clear glass or transparent jar	Must be transparent! Students need to see the coloured water level and the celery inside.
● Water	Plain tap water works perfectly. Room-temperature water is best.
● Red or blue food colouring	Use 10–15 drops for a rich, visible colour. Red is easiest to see in green leaves.
● Scissors	Adult use only! Teacher or parent cuts the bottom of the stalk at a slight angle.
● Observation worksheet + pencil/crayons	Students use this to draw and record their observations before and after the experiment.

8B — Step-by-Step Experiment Instructions

1	Fill the Glass Fill a clear glass 3/4 full with water. Add 10–15 drops of red or blue food colouring. Stir well until the colour mixes evenly throughout. Hold the glass up to the light — it should look deeply coloured!
2	Trim the Celery Ask an adult to trim the bottom of the celery stalk at a slight angle (diagonal cut) using scissors. This angled cut creates MORE surface area for the celery to absorb water — just like how we sharpen a pencil to get a better tip! Safety: Students must NOT handle scissors.
3	Check the Leaves Make sure your celery stalk has fresh, green leaves at the top. Gently touch them — they should feel firm and slightly waxy. These are the leaves that will show the colour change most clearly. If leaves are wilted, choose a fresher stalk.
4	Place in Glass Put the celery stalk cut-end DOWN into the coloured water. Make sure the stalk stands upright. You can lean it against the side of the glass. The cut end should be fully submerged in the coloured water.
5	Wait and Observe Now the fun begins! Wait and watch. Check after 30 minutes — you may already see the stem start to look slightly coloured. Good colour change in leaves appears in 2–4 hours. The BEST results come overnight!
6	Record Observations While waiting, complete the observation worksheet. Draw the celery BEFORE the experiment. After the colour change appears, draw it AFTER. Fill in the sentence strip. Discuss findings with classmates.

8C — What Will You See? (Expected Results)

After 2–4 hours (best overnight):

- The edges and tips of the celery leaves will turn the colour of the water (red or blue).
- If you cut the celery stem horizontally, you will see coloured dots — those are the tiny tubes (xylem)!
- The stem may also show coloured streaks running from bottom to top.
- The water level in the glass may drop slightly as the celery drinks it up!

◆ SECTION 9 — DISCUSSION QUESTIONS

Use these questions to guide class discussion after observing the celery experiment. Encourage all students to share — there are no wrong ideas!

Basic Recall	Q: What colour did the water become? What colour did the celery leaves become? <i>Expected: Expected answer: Water turned red/blue. Celery leaves got coloured at the edges/tips.</i>
Understanding	Q: How did the colour get from the water at the bottom all the way up to the leaves? <i>Expected: Guide students: The tiny tubes in the stem carried it up — just like a straw!</i>
Connection	Q: Do you think tulsi, neem, and other plants also have these tiny tubes inside? Why do you think so? <i>Expected: Expected: Yes, because all plants need to carry water from roots to leaves.</i>
Creative	Q: What would happen if we used TWO different colours — one on each side of the split celery stalk? <i>Expected: Fun extension: The celery would turn TWO colours — one on each side! (Try this as a home experiment.)</i>
Application	Q: Why do flower sellers keep roses and marigolds in buckets of water at the market? <i>Expected: Because the stems keep absorbing water and sending it to the petals, keeping them fresh longer.</i>

◆ SECTION 10 — OBSERVATION WORKSHEET GUIDE

Students complete this worksheet during and after the experiment.

Drawing Task:

BEFORE the Experiment Draw the celery stalk in plain water. What colour is it?	AFTER the Experiment Draw the celery stalk with coloured leaves. What colour did the leaves turn?
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Sentence Strip (Fill in the Blanks):

"The colour travelled from the _____ up through the _____ to reach the _____."

(Answer: water / coloured water — stem — leaves)

◆ SECTION 11 — KEY WORDS TO REMEMBER

■ ROOTS	Soak up water from the soil. They are underground. <i>Like a drinking straw buried in the ground!</i>
■ STEM	Carries water upward through tiny tubes. <i>Like a water highway or a bundle of straws!</i>
■ LEAVES	Receive water; stay fresh and make food for the plant. <i>Like the destination at the end of the water journey!</i>
■ ABSORB	To soak up or take in. Roots absorb water. <i>Like how a sponge absorbs water!</i>
■ OBSERVE	To look carefully and notice changes. <i>Scientists always observe before they conclude!</i>

◆ SECTION 12 — LESSON RECAP (WHAT DID WE LEARN?)

Plants Need Water Plants need water to grow strong and stay healthy! Without water, they become droopy and sad — just like Arjun's tulsi plant.	The Water Journey Roots absorb water from the soil. The stem carries it upward through tiny tubes. The leaves receive water at the top!	We Can See It! The celery experiment showed us water travel we can actually see — the coloured water travelled from bottom to top, turning the leaves!
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The Complete Water Journey:



◆ SECTION 13 — HOME EXTENSION ACTIVITY

Tonight, Look at Plants at Home!

Tonight, look at any plant at home — tulsi, coriander, rose, or money plant. Water is travelling through them right now, even as you look at them! You cannot see it, but you know it is happening — because of science!

Extension Challenge:

Try the celery experiment with white mogra (jasmine) flowers and red food colouring. Watch the white petals slowly turn pink or red! Draw what you see and bring your drawing to class tomorrow.

◆ SECTION 14 — TEACHER NOTES & DIFFERENTIATION

Common Misconceptions to Address:

X Plants make their own water.

✓ **Correction:** Plants make their own **FOOD** (using sunlight), but they get water from the soil through roots.

X Leaves absorb water from rain directly through their surface.

✓ **Correction:** While leaves can absorb a tiny amount, the primary water source is roots from soil.

X The stem is just for support.

✓ **Correction:** The stem has **TWO** jobs — support **AND** water transport through internal tubes.

Differentiation Strategies:

Struggling Learners	• Use actual plant specimens (tulsi, palak) to make roots/stem/leaves concrete. • Use a real drinking straw to act out water travelling up. • Focus only on LO1 and LO3 — name parts + do experiment. • Allow partner work for the worksheet.
On-Track Learners	• All 5 LOs as planned. • Complete the full observation worksheet. • Participate in all 5 discussion questions. • Try the home extension with mogra flowers.
Advanced Learners	• Ask: What would happen if the stem was cut in half? Would water still travel? • Research the scientific term 'xylem' for the tubes. • Try the two-colour split celery experiment at home. • Write a short paragraph: 'I am a water molecule. Describe my journey.'

◆ SECTION 15 — SUCCESS CRITERIA & ASSESSMENT

A student has successfully met the lesson goals if they can:

✓	Name the three plant parts involved in water travel (roots, stem, leaves).
✓	Explain — in their own words — how water moves from soil to leaves.
✓	Complete the celery experiment correctly with teacher guidance.
✓	Draw the BEFORE and AFTER celery sketches on the observation worksheet.
✓	Fill in the sentence strip with the correct words.
✓	Answer at least 3 of the 5 discussion questions with logical responses.

This lesson plan was generated based on the Water Travel in Plants (Celery Experiment) slide deck. For classroom use only. Not for external distribution.